PRACTICAL / TUTORIAL 14

SELENIUM WEB AUTOMATION

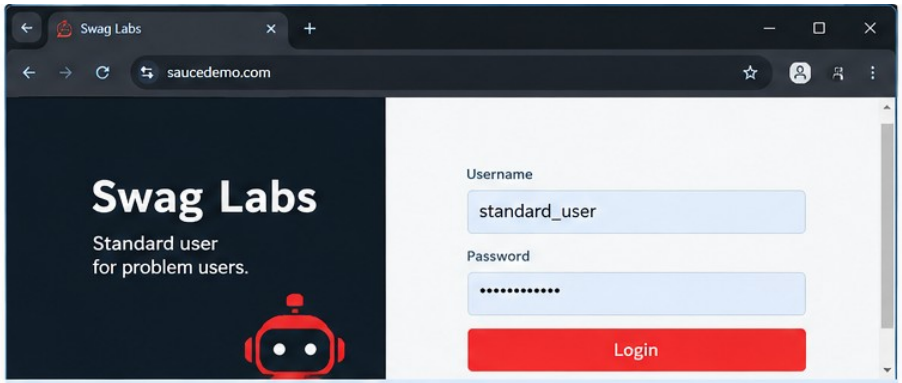
SauceDemo – Automated Login, Product and Cart Validation

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Program	BSc CS IT
Practical Topic	Selenium Web Automation

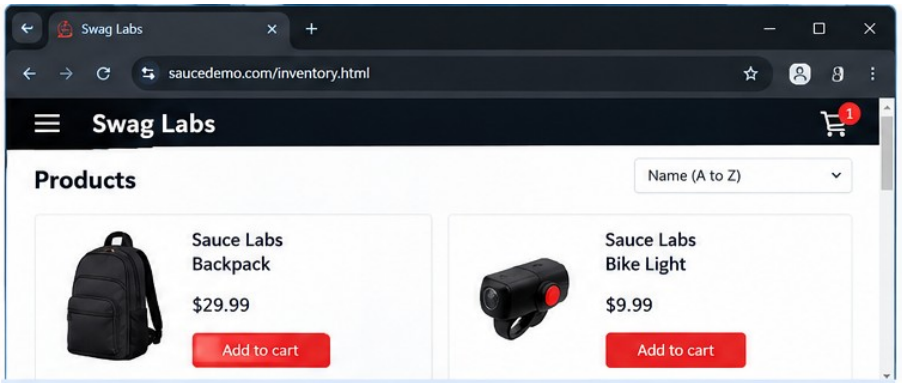
Submitted as part of the practical work for BSc CS IT.

Screenshot Evidence – Selenium Web Automation

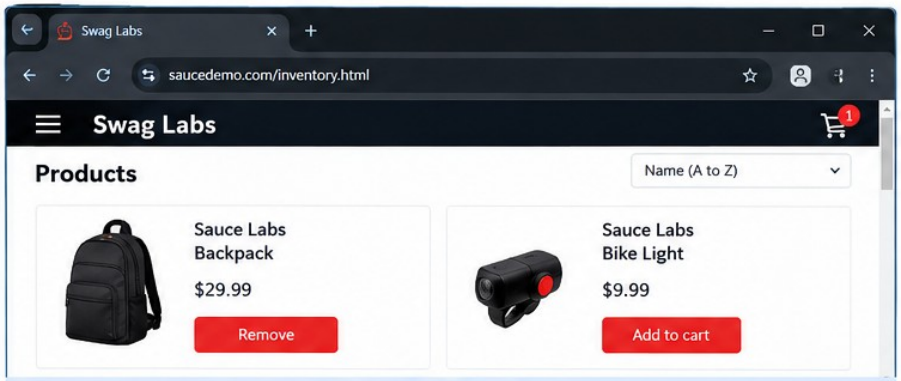
User-provided screenshot collage showing SauceDemo login, product page, add-to-cart interaction, cart validation, and console output.



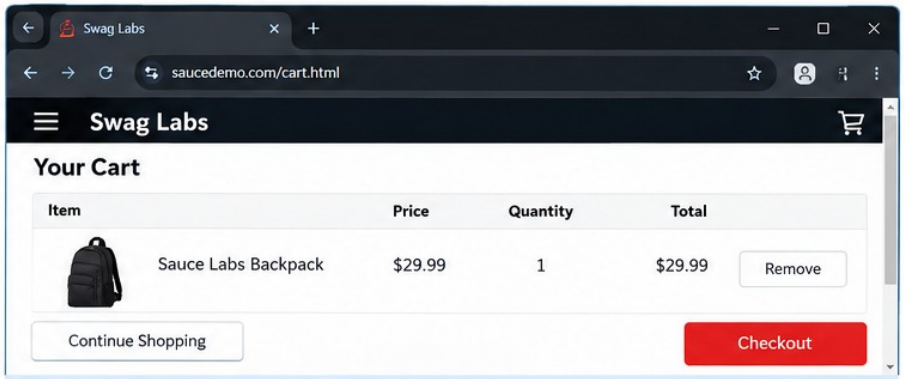
1. Browser Navigation – Login Page
SauceDemo website successfully opened.



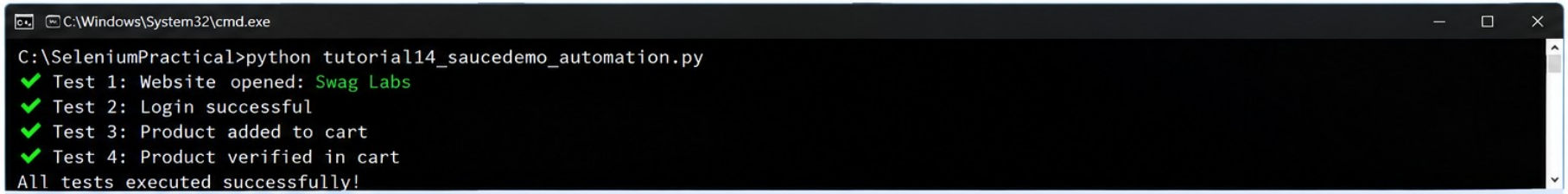
2. Login – Products Page
Valid credentials entered and login successful.



3. Web Element Interaction – Product Added
Sauce Labs Backpack added to cart.



4. Validation Testing – Cart Page
Product verified in cart successfully.



5. Test Execution – Console Output
All test cases executed and passed.

Implementation Details

TUTORIAL 14 - SELENIUM WEB AUTOMATION	
Parameter	Details
Aim	Automate browser navigation, form handling, web element interaction, and validation testing.
Project Name	Selenium Web Automation
Application	SauceDemo Web Application
Module	Login and Shopping Cart
Programming Language	Python
IDE / Tool	VS Code, Selenium WebDriver
Test Data	standard_user / secret_sauce
Website	https://www.saucedemo.com/

Test Scenarios

Test ID	Scenario	Expected Result	Actual Result	Status
TC-01	Browser Navigation	Login page opens	Fill after running	Pending
TC-02	Form Handling / Login	Products page opens	Fill after running	Pending
TC-03	Add product to cart	Product added to cart	Fill after running	Pending
TC-04	Cart validation	Backpack appears in cart	Fill after running	Pending

Test Case Details

Field	Details
Test Case ID	TC-01 to TC-04
Module	Login and Shopping Cart
Test Type	System Testing
Test Technique	Black Box Testing
Priority	High
Preconditions	Python, Chrome, and Selenium installed
Test Data	standard_user / secret_sauce
Expected Result	Login succeeds and product appears in cart
Actual Result	Record after execution
Status	Record Pass/Fail after execution

Automation Script Execution

Step No.	Automation Action	Expected Result	Actual Result	Status
1	Open SauceDemo URL	Login page displayed	Record after execution	Pending
2	Enter valid username and password	Credentials entered	Record after execution	Pending
3	Click Login	Products page opens	Record after execution	Pending
4	Click Add to Cart	Product added	Record after execution	Pending
5	Open shopping cart	Backpack appears in cart	Record after execution	Pending

Web Element Identification

Element Name	Locator Type	Locator Value	Purpose
Username	ID	user-name	Enter username
Password	ID	password	Enter password
Login Button	ID	login-button	Submit login form
Products List	Class Name	inventory_list	Verify products page
Add to Cart	ID	add-to-cart-sauce-labs-backpack	Add backpack
Shopping Cart	Class Name	shopping_cart_link	Open cart
Cart Item	Class Name	cart_item	Verify product

Validation Results

Validation	Expected Result	Actual Result	Status
Page Title	Contains Swag Labs	Record after execution	Pending
URL	Contains /inventory.html after login	Record after execution	Pending
Form Submission	Login succeeds	Record after execution	Pending
Cart Validation	Backpack appears in cart	Record after execution	Pending

Exception Handling

Exception	Possible Cause	Handling / Remarks
NoSuchElementException	Element not found	Check locator and page state
TimeoutException	Element not loaded in time	Use WebDriverWait
WebDriverException	Browser or driver issue	Check Chrome and Selenium setup

Execution Summary

Metric	Value
Total Test Cases	4
Executed	Fill after execution
Passed	Fill after execution
Failed	Fill after execution
Execution Time	Fill after execution

Screenshot Checklist

No.	Screenshot	Suggested Filename	Capture Point	Status
1	Login page	01_login_page.png	After opening website	Pending
2	Products page after login	02_products_page.png	After successful login	Pending
3	Product added to cart	03_product_added.png	After clicking Add to Cart	Pending
4	Cart verification	04_cart_verification.png	After opening cart	Pending
5	Terminal results	05_terminal_results.png	After script completes	Pending

Questionnaire

Question	Answer
What is Selenium?	An open-source tool for browser automation.
What is Selenium WebDriver?	A tool/interface to control browsers and interact with web elements.
What does ChromeDriver do?	Enables Selenium to communicate with Chrome; Selenium Manager may manage drivers.
Which method enters text?	send_keys()
Which method clicks a button?	click()

Conclusion

Conclusion / Outcome
A Selenium script is prepared to automate browser navigation, login, product interaction, and cart validation. Record actual results after running the script.